

Vision-based Static Hand Gesture Recognition using Dense-block Features and SVM classifier

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Abstract—Recognition of static hand gesture images using a vision-based technique remains a challenging task due to the variations in illumination, shape of the user's hand, complex backgrounds, and so on. Further, the current convolutional neural network (CNN) technique requires large gesture images to train a model from scratch. Therefore, in this work novel features are obtained from the dense blocks of the fine-tuned DenseNet 201 to recognize the hand gesture images accurately. The proposed feature (denoted as DenseFeat) is a combination of features from last two dense blocks of the fine-tuned DenseNet which gathers enriched information for a gesture image. In addition, to remove the redundant information in the feature vector, the dimension of the DenseFeat is reduced using the principal component analysis (PCA) approach. A support vector machine (SVM) classifier based on a linear kernel is used to recognize the test gesture images. The efficacy of the proposed method is evaluated on two benchmark datasets using both subject-independent and subject-dependent cross-validation techniques. Furthermore, the qualitative and compressive quantitative analysis of the benchmark datasets illustrates that the proposed method outperforms state-of-the-art techniques in terms of mean accuracy and computational time.

Index Terms—DenseNet, DenseFeat, hand gesture recognition, transfer learning, PCA, SVM

I. INTRODUCTION

For several decades, the vision-based hand gesture recognition (HGR) technique has become a popular research area due to the high demand for the application in real life with accurate recognition of hand gestures [1]. In this technique, the user perform gestures in front of camera with bare hands and the gestures are recognized using the computer vision techniques. The HGR systems provide the user greater freedom and are more user-friendly [2]. Some key applications using the hand gesture recognition are [3], [4] includes, sign language interpretation, clinical health, robotic control, gaming and virtual interface, etc. However, despite different novel techniques developed by the researcher, still, hand gesture recognition in a vision-based environment remains challenging due to complex background, illumination variations, inter-class similarities, and diversity in shape and size of a human hand.

Deep convolutional neural network (CNN) technique has overcome the traditional hand-crafted based approaches due to automatic feature extraction methods and ability to handle varied conditions in the datasets [5]. In addition, the automatic feature extraction technique of the CNN remove the pre-processing operations such as illumination normalization, hand

region segmentation using a skin colour filter and so on [6]. Generally, several layers such as convolutional, relu, pooling, fully connected and dropout layers are added sequentially to develop a deep CNN architecture. Moreover, training a deep CNN model from scratch with less gesture samples is a challenging task. To address the aforementioned issues, researchers employed the transfer learning method, or the extraction of deep features from the various intermediate and fully connected (FC) layers of deep CNNs [6], [7]. Several research studies [8], [9] have shown that deep CNN for learning distinguishing features and support vector machines (SVM) as a classifier outperforms deep CNN with softmax activation functions. The success of the combined CNN feature with the SVM classifier encouraged the development of a hybrid model for HGR in a vision-based environment. In the proposed technique, novel deep CNN features are obtained from the dense blocks of the fine-tuned DenseNet 201 and the gesture classes are recognized using the SVM classifier.

The major contributions of this manuscript are summarized below:

- Novel features are extracted from the dense blocks of the fine-tuned DenseNet 201 for the accurate recognition of hand gestures. The proposed feature denoted as DenseFeat is the concatenation of features from the last two dense blocks of the DenseNet 201. In particular, the DenseFeat combines the global information from receptive fields of the last two dense blocks which leverage more discriminate information for a gesture image.
- To reduce the redundant information in the DenseFeat feature dimension reduction approach using principal component analysis (PCA) technique is applied. The optimized features are more distinguishable and enable accurate recognition of gesture poses.
- Extensive experiments have been performed on two standard datasets such as MU and NUS II datasets and the proposed method outperforms the state-of-the-art methods using two cross-validation techniques in terms of accuracy and computational time.

The remainder of the paper is arranged as follows. Section II includes a review of relevant literature. Section III presents the framework of the proposed methodology for HGR. The experimental results and discussion to validate the proposed

technique is given in section IV. Section V summarizes the work.

II. RELATED WORKS

Some related and representative works on vision-based HGR technique are presented in this section. The survey is categorized into handcrafted-based approaches and CNN-based techniques.

The contour of the segmented hand gesture images is used to derive the handcrafted based features. Several novel handcrafted based feature extraction approaches [10]–[12] are proposed in the literature to recognize the gesture poses accurately. A set of geometric features from hand contour are reported by Fang *et al.* to represent a gesture pose [10]. Sparse representation of the hand contour using slope difference distribution (SDD) feature was proposed by Wang [11]. To describe a gesture image, a fusion of several image descriptors: gradient local auto-correlation, fast discrete curve transform and gabor wavelet transform were obtained from the hand gestures and the features are optimized using the PCA technique [12]. However, the aforementioned handcrafted attributes are influenced by distortion due to hand shape caused by improper hand segmentation, variations in lighting conditions, and complex environments [10].

Deep learning technique has performed well in image recognition from several years. Several CNN architectures have been reported by the researcher for the recognition of static hand gestures. A light-weighted intensive feature extraction deep network (ExtriDeNet) is reported for HGR [13]. For automatic recognition of hand gesture residual block intensity feature with a two-stage residual CNN (RBI-2RCNN) [6] is developed. A hybrid feature attention network (HyFiNet) was developed for HGR [13]. A combination of various fully connected layers attributes from two pre-trained CNNs [7] was presented for automatic HGR. A CNN network based on feature fusion was proposed [14], in which several handcrafted features from the hand contour were fused at the final FC layer to recognize a gesture pose. Although deep learning has excellent generalization capabilities, it demands a completely developed CNN architecture and large storage capacity.

III. PROPOSED METHODOLOGY

The framework of the developed HGR system is shown in the Fig. 1. The HGR system is divided into two sections: training and testing. In the training phase, first the pre-trained DenseNet 201 is fine-tuned using the training gesture samples. The novel features are then extracted from the dense blocks of fine-tuned DenseNet. The final proposed feature (denoted as DenseFeat) is the concatenation of the information from the last two dense blocks. The dimension of the proposed DenseFeat is then reduced using PCA technique. The support vector machine (SVM) model is developed using the reduced features. In the testing phase, the test gesture images are classified using the trained SVM model with the proposed feature.

A. Fine-tuning of pre-trained DenseNet 201

DenseNet uses feature reuse concept through dense connectivity among the convolutional (conv) layers in a given dense block [15]. That means the output feature maps (FM) of the present conv layer is concatenated with the FM of previous conv layer. In this way DenseNet enhances gradient flow and propagation of feature maps among conv layers within a dense block as shown in Fig. 2. The i_{th} conv layer in a dense block, receives the concatenation of FM of all the previous layers represented as $Z_0, Z_1, Z_2, \dots, Z_{i-1}$ as input:

$$Z_i = H_i[Z_0, Z_1, Z_2, \dots, Z_{i-1}] \quad (1)$$

where $[Z_0, Z_1, Z_2, \dots, Z_{i-1}]$ denotes the concatenation of FM from the previous layers 0, 1, 2, ..., $i - 1$. H_i is known as composite function. This function comprises three operations: batch normalization, ReLU, and convolution. The advantages of DenseNet are (i) it reduces the gradient vanishing problems, (ii) improve the propagation of features, (iii) inspire the reuse of features.

Fine-tuning is the process of training a new model on data from the target hand gesture dataset utilizing weights of a pre-trained trained CNN. In this work, Densenet 201 CNN model trained on ImageNet dataset are fine-tuned on the target hand gesture dataset. In this method, the output nodes in the last FC layer of the network is modified to match the total output classes in the target dataset. After that, the weights of new network is tuned on the target dataset.

B. Proposed DenseFeat feature

In this subsection, the steps to obtain proposed DenseFeat is described. The DenseNet 201 architecture consists of four dense blocks as shown in Fig. 2. In general, the initial layers of the CNN represents the low level descriptors such as lines, curves, edges etc. and the deeper layers of the CNN stores the high level abstract information of the input image [6]. The proposed feature DenseFeat is obtained from the last two dense blocks such as dense block 3 (DB3) and dense block 4 (DB4) as visualized in the Fig. 2. The output of each FM in a dense block is represented in the form of matrix as

$$K = [F_1, F_2, \dots, F_p] \quad (2)$$

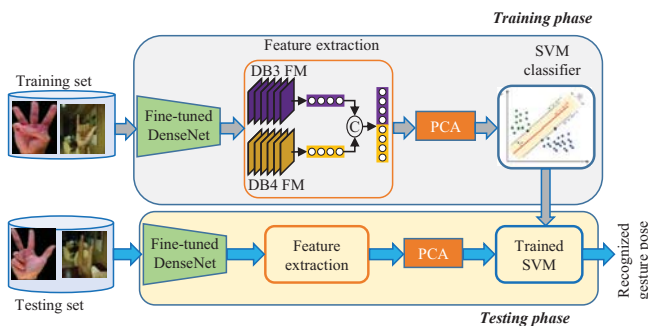


Fig. 1. The framework of the proposed HGR system. The dense block 3 and dense block 4 feature maps are represented here by DB3 and DB4 FM, respectively.

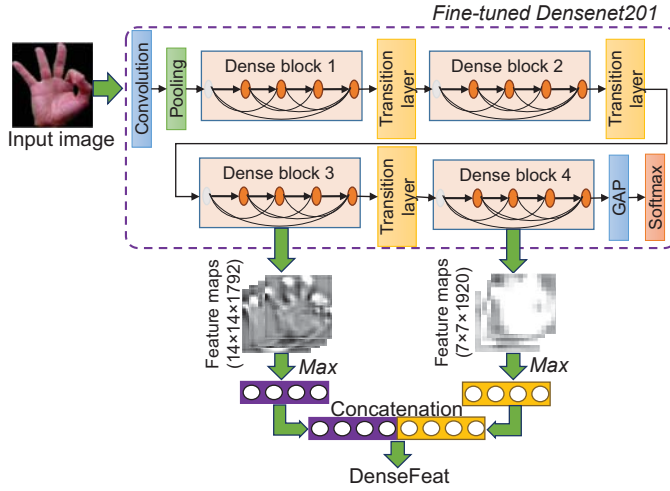


Fig. 2. Extraction of the proposed DenseFeat feature from the fine-tuned DenseNet 201 CNN.

$$[F_i]_{i=1}^p = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix} \quad (3)$$

where F and p are the learned filter wights and number of filters applied respectively. The global max pooling operation is performed on each FM to obtain the desired features for a dense block using the equation given bellow.

$$F^{max} = [F_1^{max}, F_2^{max}, \dots, F_p^{max}] \quad (4)$$

where, $[F_i^{max}]_{i=1}^p = \max(\max(F_i))$ is the maximum value of the each FM. The above features are obtained from the DB3 and DB4 which are denoted as F_{db3}^{max} and F_{db4}^{max} respectively. The concatenation of above features is the proposed feature DenseFeat (F_c) which is represented by the following equation.

$$F_c = [F_{db3}^{max}, F_{db4}^{max}] \quad (5)$$

As shown in the Fig. 2, the feature length after obtaining features from DB3 and DB4 are 1792 and 1920 respectively. Therefore, the length of the proposed feature DenseFeat is 3712.

C. PCA based feature dimension reduction

The feature dimension reduction approach is a method for reducing a higher-dimensional feature vector to a reduced dimension. In general, redundant features in higher dimensional feature vectors have low variance and are undesirable. As a result, the goal of dimension reduction is to remove redundant features in the data. The benefits of feature dimension reduction technique are as follows: (i) It decreases the amount of storage space required during training, (ii) The training algorithm has become more efficient, and (iii) eliminates unwanted features and noise from the data. As illustrated in the Fig. 1, PCA is used to decrease the dimension

TABLE I
DETAILS OF BENCH-MARKED DATASETS USED TO DETERMINE THE EFFECTIVENESS OF THE PROPOSED METHOD

Dataset name	No. of subjects	No. of gesture class	Total gesture samples	Challenges
MU [16]	5	36	2515	Illumination, hand shapes
NUS-II [17]	40	10	2000	Complex backgrounds

of DenseFeat. Before applying to the SVM classifier, the final features are normalized using Z-score normalization technique.

D. Hand gesture classification

To determine the superior recognition performance for the proposed features, three different classifiers, including artificial neural networks (ANN), K -nearest Neighbor (KNN), and support vector machines (SVM) classifiers are utilized. The performance of the individual classifier is described in the Section IV.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

Experiments are carried out in this section to validate the datasets, numerous parameter studies, and performance analyses of the proposed technique.

A. Datasets, experimental setup, and validation methods

1) *Bench-marked Datasets*: In this work, two vision-based static hand gesture datasets such as Massey University (MU) [16] and National University of Singapore (NUS)-II [17] datasets are used for the experiments. The detail description of the standard datasets is given in Table I.

2) *Experimental setup*: The experiments are performed on MATLAB 2019b platform using a 8GB graphics card. The hyper-parameters for the fine-tuning process are as follows.: maximum epoch = 20, momentum = 0.9, batch size=8 and initial learning rate = 0.001. The stochastic gradient descent with momentum (SGDM) is used to reduce the loss function.

3) *Validation methods*: The efficacy of the proposed method is evaluated using two cross validation (CV) tests such as subject independent (SI) [14] and subject dependent (SD) [13]. The SI CV test, is also known as leave-one-subject-out

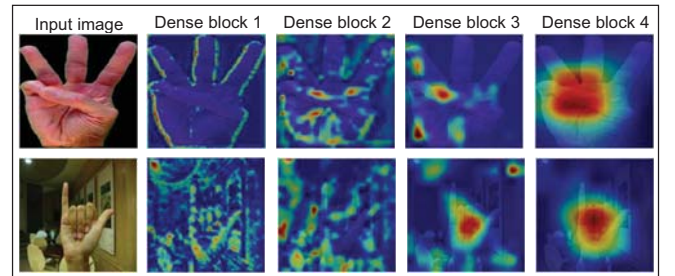


Fig. 3. Grad-CAM visualization result at each dense blocks of the fine-tuned DenseNet 201 CNN model for the given input images on the benchmark dataset. *Top*: MU dataset, *Bottom*: NUS II dataset.

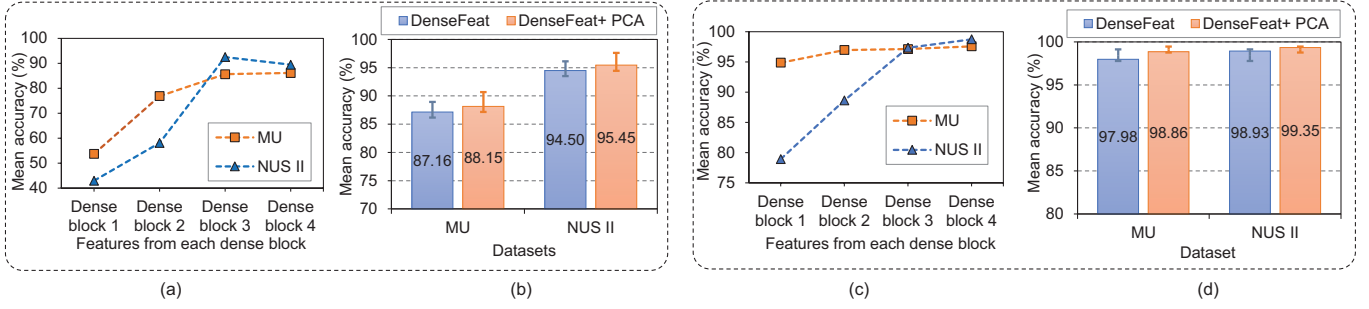


Fig. 4. Quantitative performance of proposed method using SI CV test ((a), (b)) and SD CV test ((c), (d)) on standard datasets. (a), and (c) represent results using features from each dense blocks of DenseNet 201 on SI and SD CV test respectively. (b), and (d) represent the performance comparison between DenseFeat and DenseFeat with PCA on SI and SD CV test respectively.

(LOO) CV. For the SD CV test, the gesture samples in a dataset are randomly split into training and testing sets with a ratio of 70:30. The mean accuracy is determined by repeating SD CV test ten times.

B. Evaluation of the proposed technique

1) *Qualitative analysis using grad-CAM*: The qualitative analysis of the proposed feature is achieved using the gradient-weighted class activation mapping (grad-CAM) method [18]. This technique uses the gradients of the input gesture image flowing to the last convolutional layer to generate the localization map. This heat map highlights the important parts/pixels of the image which contribute more to predict the final gesture class. Fig. 3 shows the grad-CAM results generated at each dense blocks of the fine-tuned DenseNet 201 CNN model for the input gesture image. It is observed from the generated result that the dense block 3 and dense block 4 highlight the most important regions for which the gesture pose is recognized more accurately. The recognition accuracy using the features from each dense blocks is analyzed in next subsection.

2) *Quantitative analysis*: The effectiveness of the proposed technique is observed on test gesture samples of the benchmark dataset in terms of mean accuracy. The results of the proposed method using both the CV test is shown in Fig. 4. The performance of the HGR system using proposed method is determined using four common measures such as accuracy (A), precision (P), recall (R), and F1-score (F1) [13]. The efficacy of test gesture samples utilizing the features from each dense block of the fine-tuned DenseNet 201 is represented in Fig. 4(a) and Fig. 4(c). From the above figures it is observed that the HGR performance using the features from dense block 3 and dense block 4 are superior than the first two dense blocks. This is because the first two dense blocks preserve low-level information with low variance, whereas the last two dense blocks extract high-level semantic information with high variance for an input gesture image. From this conclusion, the proposed feature is developed using the concatenation of last two dense blocks which is denoted as DenseFeat. Furthermore, the dimension of the DenseFeat is reduced using the PCA technique to eliminate the redundant information in

TABLE II
RESULTS OF PRECISION (P), RECALL (R), AND F1-SCORE (F1) ON BENCHMARKED DATASETS USING PROPOSED FEATURES

Performance measures (%)	MU dataset		NUS II dataset	
	SI test	SD test	SI test	SD test
<i>P</i>	89.31	99.28	96.46	99.68
<i>R</i>	88.06	98.16	95.14	99.04
<i>F1</i>	88.10	98.64	95.32	99.26

the DenseFeat. In this approach, the variance of DenseFeat that exceeds 99.99% is considered a redundant feature. For example in NUS II dataset using the SI CV test, the feature dimension of DenseFeat is reduced from 3712 to 1792 using the PCA technique with 99.99% variance. The performance of DenseFeat and DenseFeat with PCA using SI CV test and SD CV test is shown in Fig. 4(b) and Fig. 4(d) respectively. The performance measures such as precision, recall, and F1-score of the proposed features using both the CV test on benchmarked datasets are presented in Table II. The table shows the effectiveness of the proposed technique on above performance measures on both the dataset.

The class-wise performance comparisons on MU and NUS II datasets using SI CV test are shown in Fig. 5(a) and Fig. 5(b) respectively. It is observed from the Fig. 5(a) that the proposed technique can able to recognize the gesture poses such as '3', 'b', 'c', 'e', 'f', 'h', 'j', 'l', 'q', and 'r' in 100%. However, for the similar gesture poses like '0'-'o', '2'-'v', '6'-'w', and 'm'-'n' the recognition performance is restricted. Similarly for the NUS II complex background dataset the poses like 'b', 'f', and 'j' are most accurately recognized. However, gesture pose 'd', 'h' 'a' are misclassified due to the similar in gesture vocabulary. This implies that the representation of the aforementioned gesture postures is very identical. The Fig. 5 demonstrates that even though the proposed HGR system exhibit superior recognition accuracy in the majority classes, the overall performance is constrained due to the inter-class similarity between the gesture poses. As a result, reducing inter-class similarity in gesture vocabulary is necessary for successful hand gesture recognition.

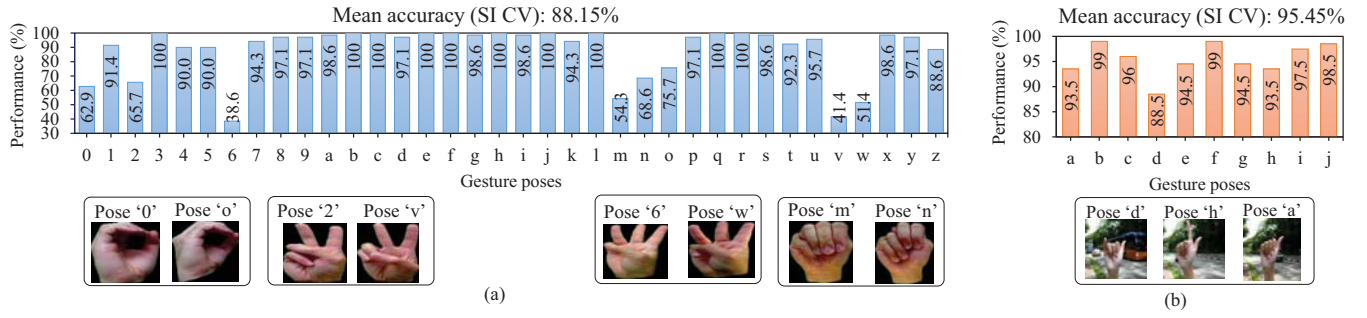


Fig. 5. Class-wise performance comparison of DenseFeat with PCA using SI CV test on both the datasets. (a) MU dataset, (b) NUS II dataset.

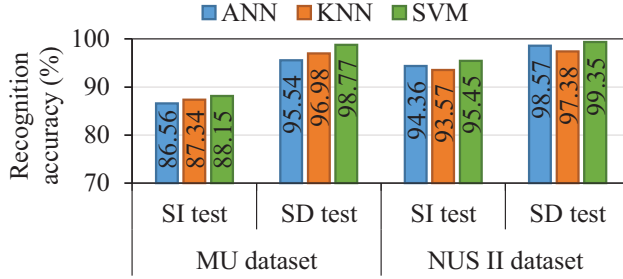


Fig. 6. Performance of proposed technique using individual classifier.

C. Classification performance using individual classifiers

The proposed feature DenseFeat, after feature optimization, is given into the classifiers ANN, KNN, and SVM to recognize the hand gestures. For ANN, the training algorithm uses multi-layer perception with a backpropagation algorithm. The "Log-sigmoid" activation function is considered for the both hidden and output layers. KNN is a supervised machine learning algorithm that recognizes a new data point among its K nearest neighbors as belonging to the majority class. The Euclidean distance parameter is employed in this study to determine the nearest neighbors. Experimentally, the K value is varied from 1-10 and the highest recognition accuracy is obtained for the K value as 5. A liner kernel-based multi-class SVM is used to classify hand motions using "one-against-all" (OAA) method [19]. For the training data having M output classes, M binary SVM classifiers are generated using the OAA technique. The performance of the individual classifier is shown in shown in Fig. 6. The results shows that the recognition accuracy of proposed feature using SVM classifier is superior than ANN and KNN.

1) *Performance comparison of proposed feature with features from other state-of-the-art CNNs*: This subsection compares the effectiveness of the proposed DenseFeat with the fusion of features from the last two blocks of the state-of-the-art CNNs. For VGG 16 network, the features are extracted from the 'conv4_3' and 'conv5_3' layers, and the test performance is obtained using the SVM classifier. Similarly, for ResNet 18 and ResNet 50 CNN models, the features are obtained from the last two residual blocks. The particular feature extraction

layers for both the ResNet models are presented in Table III. For the proposed DenseFeat, the feature extraction layers for last two dense blocks are 'conv4_block48_concat' and 'conv5_block32_concat'. The extracted features from both layers are concatenated and used to train the SVM model. The performance is evaluated using the trained SVM on the test gesture samples, which is shown in Table III. The result indicates that the features from DenseNet 201, which is the proposed DenseFeat provide superior performance on both the benchmarked datasets.

2) *Computational time*: The average inference time required to predict a gesture pose using the proposed DenseFeat with PCA, ExtriDeNet [22], and HyFiNet [13] are 0.81s, 1.30s, and 6.4s respectively. This demonstrates that the proposed method can able to recognize the gesture pose more accurately with less time compared to earlier reported techniques.

D. Comparison with earlier techniques on bench-marked datasets

Table IV shows the comparison of recognition accuracy of the proposed technique versus state-of-the-art approaches on both datasets. The result indicates that the proposed method outperforms earlier reported methods, achieving a mean accuracy of 88.15 % and 95.45 % using the SI CV test on the MU and NUS II datasets, respectively. Similarly for SD CV test, the proposed technique achieves higher mean accuracy than the previous methods. Furthermore, it can also be observed from the tabulation result that as the SD CV test is a user-

TABLE III
COMPARISON RESULTS OF PROPOSED DENSEFEAT AGAINST THE FUSION OF GLOBAL FEATURES FROM DIFFERENT BLOCKS OF STATE-OF-THE-ART CNNs USING SI CV TEST

CNN models	Feature extraction layers	MU dataset	NUS II dataset
		Mean accuracy (%)	Mean accuracy (%)
VGG 16 [20]	conv4_3, and conv5_3	84.85	91.89
ResNet 18 [21]	res4b and res5b	84.06	93.26
ResNet 50 [21]	add_13, and add_16	85.77	93.68
DenseNet 201 [15]	conv4_block48_concat, and conv5_block32_concat	87.16	94.50

The bold letter values indicate the proposed DenseFeat results.

TABLE IV
COMPARISON RESULTS OF PROPOSED METHOD WITH STATE-OF-THE-ART
TECHNIQUES ON BOTH THE STANDARD DATASETS

Dataset	SI test		SD test	
	Methods	Accuracy	Methods	Accuracy
MU	CNN [14]	73.86	HyFiNet [13]	87.13
	HyFiNet [13]	77.44	GB-ZM [23]	97.09
	FCNN [14]	84.02	GB-HU [23]	97.63
	ExtriDeNet [22]	84.28		
	Fusion [AlexNet, VGG16] [7]	87.83	DenseFeat + PCA	98.77
NUS II	DenseFeat + PCA	88.15		
	ExtriDeNet [22]	61.49	ExtriDeNet [22]	98.75
	Fusion [AlexNet, VGG16] [7]	92.65	HyFiNet [13]	97.78
	DenseFeat + PCA	95.45	DenseFeat + PCA	99.35

biased CV approach, its performance is often superior to the SI CV test. The proposed technique shows superior performance in both the datasets due to the following reasons. (i) The proposed features DenseFeat contains the combination of high-level semantic information from the last two dense blocks of the DenseNet 201 which may preserve better discriminate information for the gesture pose. (ii) After removing the redundant information from the DenseFeat using the PCA technique the reduced feature represents distinguishable features for a given input gesture pose.

V. CONCLUSIONS

In this work, a novel feature known as DenseFeat is proposed for accurate recognition of vision-based static hand gestures. The proposed system does not require any pre-processing operations such as illumination normalization, skin colour based filter for hand segmentation, etc. The DenseFeat focuses on the global features from the receptive fields of both the dense blocks which acquires the discriminate information for the input gesture pose. In addition, the redundant information in the DenseFeat is eliminated using the PCA technique. Various qualitative and quantitative analysis proves that the proposed method provides superior performance compared to other earlier reported methods on both the standard datasets in terms of recognition accuracy and computational time. In future work, some discriminate feature extraction modules can be added to the present HGR system to reduce the similar gesture class-specific behavior between the hand gestures.

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